UNIVERSIDAD DE GRANADA

Facultad de Ciencias

GRADO EN FÍSICA

Práctica de Complejos

Quantum Chaos and the Statistics of Energy Levels

María de los Ángeles Lara
Pablo Orellana

05/2026

Abstract

bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla bla

Índice

1 Introduction

The study of complex systems allows to explore the transition between the classical order and the statistichal manifestations in the quantum regime. As in classical mechanics chaos isdefined as the exponential sensitivity to initial conditions, the linearity in the Schrödinger equation does not allow for a direct translation of this definition to quantum mechanics. However, the phenomena known as 'Quantum Chaos' is manifested throughout the statistichal properties of the energy spectrum of the system.

The nuclei in this project is the Bohigas-Giannoni-Schmit (BGS) conjecture, which states that the energy levels of a quantum system whose classical counterpart is chaotic will exhibit the same statistical properties as the eigenvalues of a random matrix ensemble.

To validate this hipothesis, we'll investigate the distribution of spacing between adjacent energy levels in two systems (s_i): one integrable and one chaotic.

The first one is the two-dimensional **rectangular billiard**, which is an **integrable system**. In this system, the random matrix theory (RMT) predicts that the distribution of level spacings follows a Poisson distribution, which is given by:

$$P(s) = e^{-s} \tag{1.1}$$

The second system is the **stadium billiard**, which is a **chaotic system**. In this case, RMT predicts that the distribution of level spacings follows the Wigner-Dyson distribution, which is given by:

$$P(s) = \frac{\pi}{2} s e^{-\frac{\pi}{4}s^2} \tag{1.2}$$

where s is the normalized spacing between adjacent energy levels in both cases.

To integrate the time independent Schrödinger equation for these systems, we will use the **Finite Difference Method** (FDM): this method allows us to discretize the spatial domain and solve for the energy eigenvalues and eigenfunctions of the system.

2 Methods

To be able to solve the Schrödinger's Equation $H\psi = E\psi$, we'll first need to discretize the space and then solve the resulting eigenvalue problem. We'll need to impose the Dirichlet conditions to the domain, which means that the wavefunction ψ must vanish at the boundaries of the billiard.

$$\psi|_{\partial\Omega} = 0 \tag{2.1}$$

All the methods used in this project are implemented in Python, and the code is available in the GitHub repository https://github.com/1313ss/PCS-Project-.

To explain the followed procedure, we'll first introduce the Mascara method, which is used to discretize the system, and then the Finite Difference Method, which is used to solve the resulting eigenvalue problem.

2.1 Discretisation of the system: Mascara Method

2.2 Finite Difference Method

3 Results

4 Conclusions

Referencias